

Group Assignment: From Analysis to Design

Stomach Saver – Online Food Ordering System

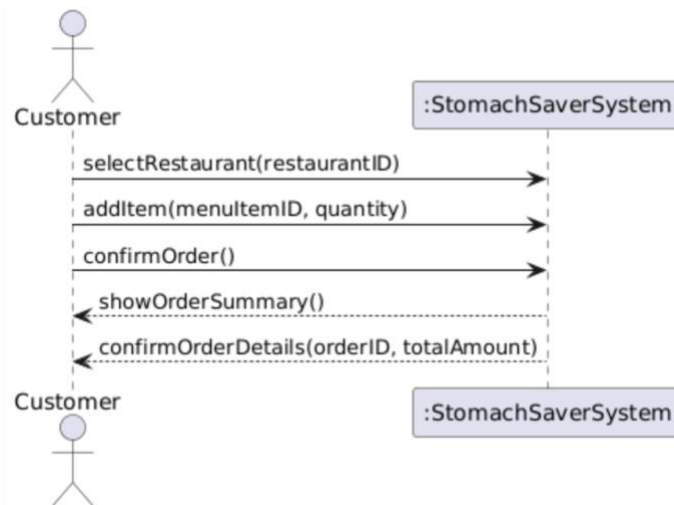
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Part 1 – System Sequence Diagram (SSD)

A System Sequence Diagram (SSD) is a diagram that shows how an external actor interacts with the system as a whole. The system is treated as a black box, meaning that internal objects and implementation details are not shown. SSDs focus on what the system does rather than how it is implemented internally. They are derived from use case descriptions and help move from analysis to design.

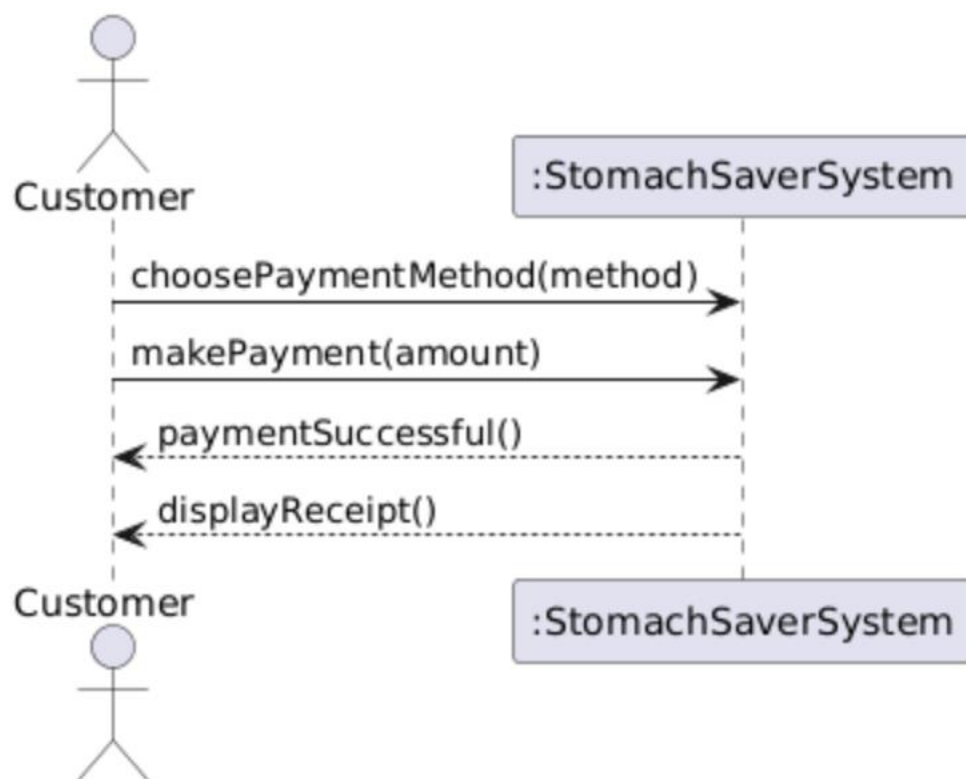
SSD 1 – Place Food Order (Stomach Saver)

This System Sequence Diagram describes how a customer places a food order using the Stomach Saver system. The customer selects a restaurant, adds menu items to the order, and confirms the order. The system then responds by displaying an order summary and confirming the order details.



SSD 2 – Make Payment (Stomach Saver)

This diagram illustrates how a customer completes the payment process. The customer selects a payment method and makes a payment. The system processes the payment and confirms it by displaying a receipt.



Part 2 – Operation Contracts (OC)

Operation Contracts describe how the internal state of the system changes after an operation is executed. They focus on object creation, associations, and attribute changes.

Operation Contract – confirmOrder()

Preconditions:

A Customer instance exists.

An Order instance exists and is not yet confirmed.

Postconditions:

The Order was confirmed.

The Order status was set to Confirmed.

The Order was linked to the Customer.

The total amount was calculated.

Operation Contract – makePayment(amount)

Preconditions:

A confirmed Order exists.

A Customer exists.

Postconditions:

A Payment instance was created.

The Payment was linked to the Order.

The Order status was updated to Paid.

Part 3 – Sequence Diagram

A Sequence Diagram shows how objects communicate with each other over time and is used to describe internal system behavior.

